DATA STRUCTURE & ALGORITHM

LAB 7

TASK 1:-

main.cpp Run ▶ Output

```cpp
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* prev;
    Node* next;
};
int main() {
    Node* first = new Node();
    Node* second = new Node();
    Node* third = new Node();
    Node* fourth = new Node();
    Node* fifth = new Node();
    first->data = 10;
    second->data = 20;
    third->data = 30;
    fourth->data = 40;
    fifth->data = 50;
    first->prev = nullptr;
    first->next = second;
    second->prev = first;
    second->next = third;
    third->prev = second;
    third->next = fourth;
```

```
Student Roll Numbers: 5 10 20 25 30 40 50 60
```

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main.cpp Run ▶ Output

```cpp
    third->next = fourth;
    fourth->prev = third;
    fourth->next = fifth;
    fifth->prev = fourth;
    fifth->next = nullptr;
    Node* head = first;
    Node* tail = fifth;
    Node* newNode = new Node();
    newNode->data = 5;
    newNode->prev = nullptr;
    newNode->next = head;
    head->prev = newNode;
    head = newNode;
    Node* temp = head;
    while (temp != nullptr && temp->data != 20)
    {
        temp = temp->next;
    }
    if (temp != nullptr)
    {
        Node* newNode = new Node();
        newNode->data = 25;
        newNode->prev = temp;
        newNode->next = temp->next;
        temp->next->prev = newNode;
```

```
Student Roll Numbers: 5 10 20 25 30 40 50 60
```

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main.cpp Run ▶ Output

```
    }
    if (temp != nullptr)
    {
        Node* newNode = new Node();
        newNode->data = 25;
        newNode->prev = temp;
        newNode->next = temp->next;
        temp->next->prev = newNode;
        temp->next = newNode;
    }
    Node* endNode = new Node();
    endNode->data = 60;
    endNode->next = nullptr;
    endNode->prev = tail;
    tail->next = endNode;
    tail = endNode;
    cout << "Student Roll Numbers: ";
    temp = head;
    while (temp != nullptr)
    {
        cout << temp->data << " ";
        temp = temp->next;
    }
    return 0;
}
```

```
Student Roll Numbers: 5 10 20 25 30 40 50 60
```

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TASK 2:-

main.cpp Run ▶ Output

```
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* prev;
    Node* next;
};
int main() {
    Node* first = new Node();
    Node* second = new Node();
    Node* third = new Node();
    Node* fourth = new Node();
    Node* fifth = new Node();
    first->data = 101;
    second->data = 102;
    third->data = 103;
    fourth->data = 104;
    fifth->data = 105;
    first->prev = nullptr;
    first->next = second;
    second->prev = first;
    second->next = third;
    third->prev = second;
    third->next = fourth;
```

```
Product IDs: 101 102 103 1035 105
```

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main.cpp Run ▶ Output

```
    fourth->prev = third;
    fourth->next = fifth;
    fifth->prev = fourth;
    fifth->next = nullptr;
    Node* head = first;
    Node* tail = fifth;
    Node* temp;
    Node* newNode = new Node();
    newNode->data = 100;
    newNode->prev = nullptr;
    newNode->next = head;
    head->prev = newNode;
    head = newNode;
    temp = head;
    while (temp != nullptr && temp->data != 103) {
        temp = temp->next;
    }
    if (temp != nullptr) {
        Node* newNode = new Node();
        newNode->data = 1035;
        newNode->prev = temp;
        newNode->next = temp->next;
        temp->next->prev = newNode;
        temp->next = newNode;
```

```
Product IDs: 101 102 103 1035 105
```

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main.cpp Run ▶ Output

```
    Node* endNode = new Node();
    endNode->data = 106;
    endNode->next = nullptr;
    endNode->prev = tail;
    tail->next = endNode;
    tail = endNode;
    temp = head;
    head = head->next;
    head->prev = nullptr;
    delete temp;    temp = head;
    while (temp != nullptr && temp->data != 104) {
        temp = temp->next;    }
    if (temp != nullptr) {
        temp->prev->next = temp->next;
        temp->next->prev = temp->prev;
        delete temp;    }
    temp = tail;
    tail = tail->prev;
    tail->next = nullptr;
    delete temp;
    cout << "Product IDs: ";
    temp = head;
    while (temp != nullptr) {
        cout << temp->data << " ";
        temp = temp->next;    }    return 0;}
```

```
Product IDs: 101 102 103 1035 105
```

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